

Md Raihan Khan

Gilatala South Para, Shiromoni, Khan Jahan Ali, Khulna 9204, Bangladesh
(+880) 1879-439898 · kraihan918@gmail.com · <https://kraihan.github.io/> · scholar.google.com/citations?user=E3iFEuUAAAAJ

EDUCATION

Khulna University of Engineering and Technology Khulna, Bangladesh
Master of Science in Electrical and Electronic Engineering Expected December 2026
Thesis: Local–Global Feature Integration in Neural Signals Using Dual-Path Deep Learning Models
Advisor: Dr. Mohiuddin Ahmad, Professor of Electrical and Electronic Engineering

Khulna University of Engineering and Technology Khulna, Bangladesh
Bachelor of Science in Electrical and Electronic Engineering January 2019 – March 2024
Cumulative Grade Point Average: 3.71 / 4.00, ranked 24 of 149 (top 16%);
Thesis: Assessing Mental Stress Using Deep Learning Methods from EEG Signals
Advisor: Dr. Mohiuddin Ahmad, Professor of Electrical and Electronic Engineering

RESEARCH INTERESTS

- One- and few-step generative modeling: closed-form flow and transport formulations that collapse iterative sampling into a single network evaluation
- Image restoration and exposure correction under real-world degradation, formulated through principled optimal transport rather than heuristic reconstruction losses
- Geometry-aware deep learning on Riemannian and symmetric positive definite manifolds: equivariant architectures, geodesic operators, and numerically stable matrix-function approximation
- Structured representation learning for neural and biomedical signals, including EEG-based affective and cognitive state decoding

RESEARCH EXPERIENCE

Graduate Thesis Researcher November 2024 – Present
Department of Electrical and Electronic Engineering, Khulna University of Engineering and Technology
Advisor: Dr. Mohiuddin Ahmad

- Derived a closed-form flow formulation that collapses iterative diffusion sampling into a single network evaluation, removing the multi-step solver required by conventional flow matching; implemented the complete reference training codebase with stabilizers for the degenerate-path regime.
- Proposed Equivariant Geodesic Networks, an end-to-end classifier composed of equivariant mapping, Riemannian pooling, geometric bias, and geodesic attention layers that operates directly on the symmetric positive definite manifold instead of projecting to Euclidean space.
- Derived orthogonal-polynomial approximations for matrix logarithm normalization in global covariance pooling, reducing computational cost and enabling GPU execution of a previously CPU-bound operation.
- Designed AutoLumNet, a bi-branch exposure-aware network performing single-shot correction of both under- and over-exposed images; reformulated the correction step as monotone optimal transport, attaining an average MSEC score of 23.75.
- Developed DAFNet, integrating a local two-dimensional convolutional branch with channel attention and a global graph attention transformer with covariance pooling, achieving state-of-the-art accuracy across multiple EEG emotion recognition benchmarks.

Research Collaborator May 2024 – July 2024
Department of Electrical and Electronic Engineering, Khulna University of Engineering and Technology
Supervisor: Dr. Md. Mahbub Hasan

- Constructed the first EEG auditory-evoked-potential dataset for imagined Bengali language tasks using a low-cost OpenBCI acquisition system.
- Trained and evaluated convolutional models for Bengali vowel and word recognition; resulting paper received the IEEE Best Paper Award.

Undergraduate Thesis Researcher

January 2023 – March 2024

Department of Electrical and Electronic Engineering, Khulna University of Engineering and Technology

Advisor: Dr. Mohiuddin Ahmad

- Established that phase-locking value and cross-correlation features perform best with larger kernels, whereas wavelet energy and differential entropy favor smaller kernels, in three-dimensional convolutional models for EEG emotion recognition.
- Built the complete preprocessing, class-balancing, and evaluation pipeline on constrained hardware without GPU access.

PUBLICATIONS

Refereed Journal Articles

Md Raihan Khan, Airin Akter Tania, Tanjum Arifen Bushra, Jahanara Pritha, and Mohiuddin Ahmad, “DAFNet: A dual-path attention fusion network for EEG emotion recognition via CNN and graph-based global modeling,” *Array (Elsevier)*, 2025.

Md Raihan Khan, Airin Akter Tania, and Mohiuddin Ahmad, “A comparative study of time–frequency features based spatio-temporal analysis with varying multiscale kernels for emotion recognition from EEG,” *Biomedical Signal Processing and Control (Elsevier)*, 2025.

Abdullah Al Ahad, Bibhakar Das, **Md Raihan Khan**, Nitol Saha, Abu Zahid, and Mohiuddin Ahmad, “Multiclass liver disease prediction with adaptive data preprocessing and ensemble modeling,” *Results in Engineering (Elsevier)*, 2024.

Refereed Conference Proceedings

Md Rifat Ur Rahman, **Md Raihan Khan**, Md Sakib Hossain Shovon, Pietro Liò, and Mohammad Ali Moni, “Orthogonal polynomial approximation for matrix log normalization in global covariance pooling,” *37th British Machine Vision Conference (BMVC)*, accepted, 2026.

Tithi Das, **Md Raihan Khan**, and Md Mahbub Hasan, “Recognition of Bengali vowels from auditory evoked potentials using CNN,” *IEEE International Conference on Signal Processing, Information, Communication and Systems — Best Paper Award*, 2024.

Airin Akter Tania, **Md Raihan Khan**, and Mohiuddin Ahmad, “Custom dataset-driven unsupervised low-light image enhancement using 2D CNN,” *IEEE International Conference on Quantum Photonics, Artificial Intelligence, and Networking*, 2025.

Md Raihan Khan and Mohiuddin Ahmad, “Mental stress detection from EEG signals using comparative analysis of random forest and recurrent neural network,” *IEEE International Conference on Advances in Computing, Communication, Electrical, and Smart Systems*, 2024.

Rabita Hasan, Sheikh Md Rabiul Islam, and **Md Raihan Khan**, “Machine learning techniques for brain stroke analysis and prediction,” *IEEE International Conference on Signal Processing, Information, Communication and Systems*, 2024.

Md Raihan Khan, Abdullah Al Ahad, Airin Akter Tania, Tithi Das, and Bibhakar Das, “Smart classroom automation: A fusion of AI with voice, gesture, and face recognition attendance system,” *IEEE International Conference on Advances in Computing, Communication, Electrical, and Smart Systems*, 2024.

MANUSCRIPTS UNDER SUBMISSION

Airin Akter Tania, **Md Raihan Khan**, and Mohiuddin Ahmad, “AutoLumNet: Monotone optimal transport for single-shot exposure correction,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, under review, 2026.

Md Raihan Khan and Airin Akter Tania, “Equivariant geodesic networks: End-to-end classification on the SPD manifold,” *Forty-First AAAI Conference on Artificial Intelligence*, under review, 2026.

Susmita Saha Lagna and **Md Raihan Khan**, “MCFN: A lightweight multi-path compound fusion network for deepfake face detection,” *Neural Computing and Applications (Springer)*, under review, 2026.

Md Rifat Ur Rahman, **Md Raihan Khan**, Resadul Islam, and Mahfuja Akter “Trigonometric B-spline convolutional KAN for texture classification,” *Eighteenth Asian Conference on Computer Vision*, under review, 2026.

Airin Akter Tania and **Md Raihan Khan**, “Difficulty-calibrated interpolation paths for conditional flow matching,” *Third IEEE International Conference on Computing, Applications and Systems*, under review, 2026.

SOFTWARE AND OPEN-SOURCE CONTRIBUTIONS

- egnlb** — Equivariant Geodesic Networks for the SPD manifold 2026
- Authored and released an open-source Python package on the Python Package Index implementing the complete EGN architecture, including equivariant mapping, Riemannian pooling, geometric bias, and geodesic attention layers, with a stable matrix-logarithm backward pass.
 - Distributed under pip install `egnlb`; source at github.com/kraihan/EGN.

TEACHING EXPERIENCE

- Lecturer** August 2024 – Present
Department of Electrical and Electronic Engineering, North Western University, Khulna, Bangladesh
- Digital Image Processing (EEE-4255): deliver lectures on core theory, conduct laboratory sessions, design assignments, and supervise student projects in computer vision applications.
 - Digital Signal Processing (EEE-4115): teach theoretical foundations and design assignments centered on biomedical signal analysis, including the discrete and fast Fourier transforms.
 - Sessional on Digital Signal Processing (EEE-4116): conduct laboratory sessions and supervise student projects in EEG signal processing.
 - Basic Electrical Circuits (EEE-1221): teach fundamental laws, network theorems, and circuit analysis, and develop the accompanying assignment sequence.
 - Supervise three undergraduate thesis students from problem formulation through manuscript submission, teaching programming, classical machine learning, deep learning, and signal processing.

ACADEMIC SERVICE

- Batch Advisor, third-year cohort, Department of EEE, North Western University 2025 – Present
Course Coordinator, Department of EEE, North Western University 2024 – Present
Co-founder, departmental research group, North Western University 2024 – Present
Vice President, KUET Mathematics Club February 2023 – February 2024
Ambassador, International Youth Math Challenge — mathematics outreach in Bangladesh 2021

HONORS AND AWARDS

- Honorable Mention, International Mathematics Competition, Bulgaria — 589 participants worldwide 2021
First participant from Bangladesh, 27th International Mathematics Competition 2020
Gold Medal, International Youth Math Challenge — top 1% of 15,000 competitors 2020, 2021
Excellence Award, IYMC Ambassador Programme — fifth among 580 ambassadors 2021
Best Paper Award, IEEE Conference on Signal Processing, Information and Communication Systems 2024

Silver and Bronze Medals, International Astronomy and Astrophysics Competition	2022, 2021
Best Project Award, Electrical and Electronic Project Design Showcase, KUET	2023
University Dean's Scholarship, KUET — second and fourth academic years	2021, 2024
University Merit Scholarship, KUET — awarded seven times for departmental ranking	2020 – 2024

TECHNICAL PROFICIENCIES AND LANGUAGES

Programming: Python, C++, MATLAB

Frameworks and libraries: PyTorch, TensorFlow, Keras, scikit-learn, NumPy, pandas, SciPy, OpenCV, MNE, EEGLAB, Matplotlib, Seaborn, Flask

Tools: Git and GitHub, LaTeX, Linux

Methods: Riemannian and SPD-manifold learning, optimal transport, flow matching and diffusion models, graph neural networks, EEG signal processing, low-level computer vision

Languages: Bengali (native), English (professional working proficiency)

REFERENCES

Dr. Mohiuddin Ahmad

Professor, Department of Electrical and Electronic Engineering
 Khulna University of Engineering and Technology, Khulna 9203, Bangladesh
 Telephone: (+880) 1556-537660 · Email: ahmad@eee.kuet.ac.bd
Relationship: undergraduate and graduate thesis advisor

Dr. Kalyan Kumar Halder

Professor, Department of Electrical and Electronic Engineering
 Khulna University of Engineering and Technology, Khulna 9203, Bangladesh
 Telephone: (+880) 1710-752236 · Email: kalyan@eee.kuet.ac.bd
Relationship: course instructor and mentor for International Mathematics Competition preparation